FOR MORE WORKBOOKS; (Librarian) 0703181364

SCIENCE SCHOLARS ACADEMY

HALOGEN COMPOUNDS: (ALKYLHALIDES)!

Course outline:

(1). Introduction {functional group, General formula, classes, sources, occurrence, (types{formats}/etc)}

(2). Structure & Nomenclature.

(3). Isomerism

(4). Physical properties.

(5). Laboratory Preparations. {of alkylhalides}

(6). Chemical properties

(7). Identification of the functional group

(8). Applications/Uses/Roles

In daily life.

END

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Introduction:

Halogen Compounds → ALKYL HALIDES ✓
Halogen Compounds → ARYL HALIDES ✗ (Reserved for Aromaticity.)

Recall:

Halogens → Group VII (17) ✓
$\{Cl_2, Br_2, I_2\}$ + other elements = Halogen Compound

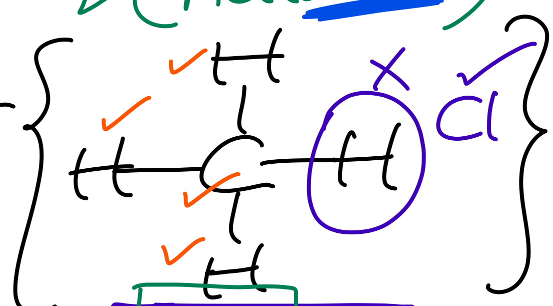
✓ Cl_2 → Chloride
Br_2 → Bromide
I_2 → Iodide

FORM

(halogens) → (halides)

For example:

$Cl_2 + CH_4 \xrightarrow[\text{light}]{\text{u.v}}$



CH_3Cl

$\underbrace{Cl_2}$

$\boxed{Na}Cl$ → sodium chloride ✓

$\boxed{CH_3}Br$

CH₃ → Methyl group

→ ~~Methyl chloride~~ ✗

$\boxed{\text{IUPAC}}$ → A $\boxed{\text{Chloro}}$ group

$H_3\boxed{C-Br}$, $H_3\boxed{C-Cl}$, $H_3\boxed{C-I}$

→ Bromo group. → Chloro group. → Iodo group.

Collectively called halo groups.

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NOT IUPAC. { Methyl chloride CH_3Cl
Ethyl bromide $\boxed{CH_3CH_2}Br$ }

{Alkyl groups} + halides → {Alkyl halides}

Meaning → Most alkyl halides have a structure $\boxed{R}\boxed{X}$ → $-Cl$, $-Br$, $-I$

Only one atom substituting!

Hence

$\boxed{\text{Mono}}$substituted Alkyl halides!

{Alkyl group}

Methyl group.

Ethyl group.

$\boxed{\text{etc}}$

Now: what if we go beyond $\boxed{\text{Mono}}$ — substitution.

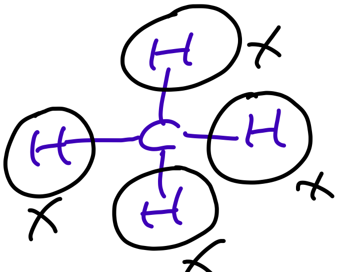
CH_4 ① → CH_3Cl → Monohalide = halide ($1x = x$)

② → CH_2Cl_2 → dihalide

③ → $CHCl_3$ → trihalide

④ → CCl_4 → tetrahalide

Poly-halides



✳ In the above case → {~~halide~~ = chloride}

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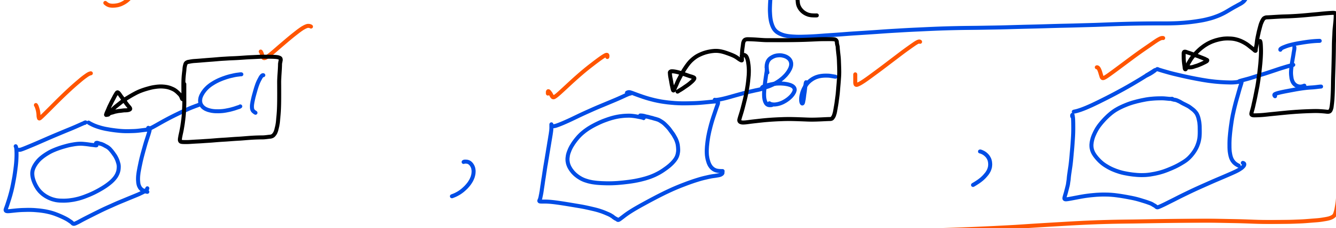
WHAT IF? → In RX

R is a phenyl group → Benzene Ring acting as a side chain.

CH_3Br

i.e



Are they still alkyl halides? NO

* → They are aryl halides

→ They are Aromatic cpds

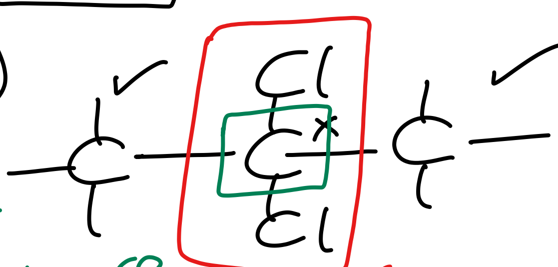
* About dihalides → $2X_r$ → X_2

They can occupy the same carbon.

OR

different carbons!

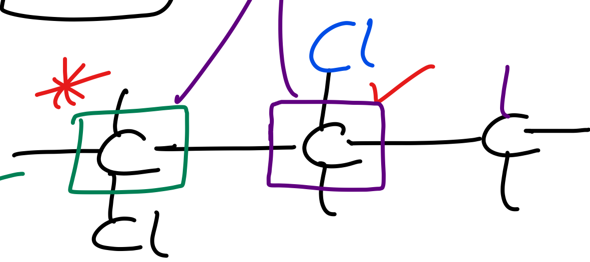
Eg



Condense → $CH_3CCl_2CH_3$

E.g

Adjacent



Condense → $CH_2ClCHClCH_3$

Geminal → from Latin word Gemini means → twins (paired)

GEMinal dihalide

Vicinal dihalide

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only Monohalides (halides)

MOST IMPORTANTLY: → Classes of alkylhalides!

Qn: $\boxed{\text{who}}$ determines the class of alkylhalides?

→ carbon having the halogen atom; i.e C—Cl, C—Br, C—I

Qn: How?

→ when that carbon atom has;

attached to it

(1). 3/2 hydrogens → Primary alkyl halide (1^{o} or 1^{ry})

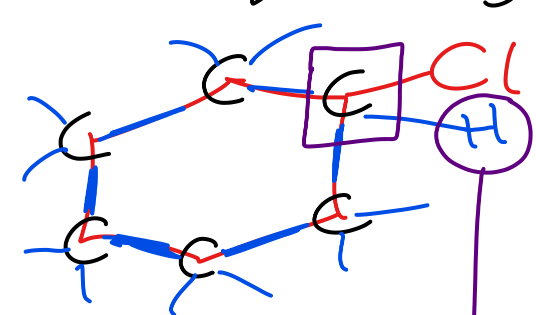
a.t.i

(2). $\boxed{1}$ hydrogen → Secondary alkyl halide ↳ 2^{o}/2^{ry}

a.t.i

(3). 0 hydrogen → terti$\boxed{\text{ary}}$ alkyl halide ↳ 3^{o}/3^{ry}

(No hydrogen)



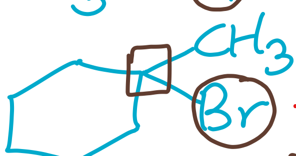
→ Secondary

CH_3Cl

→ H_3C—Cl → Primary

(1). CH_3CH_2Cl P

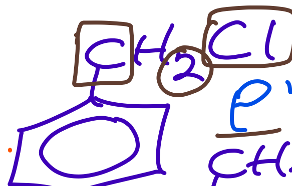
(2).



T

(3). CH_3CHCH_3 with I below — S

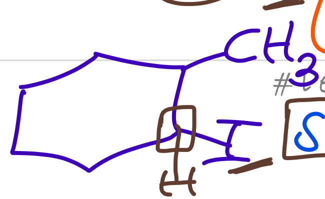
(4).



P

(5). CH_3CCH_3 with CH_3 above and Br below — T

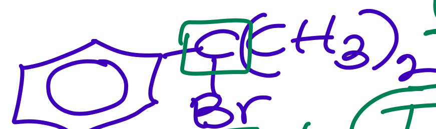
(6).



S

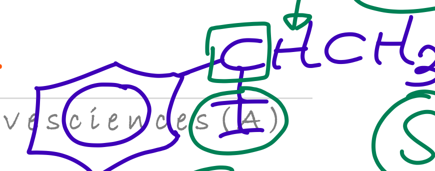
(7). $CH_3CH_2CH_2Cl$ P

(8).



T

(9).



S

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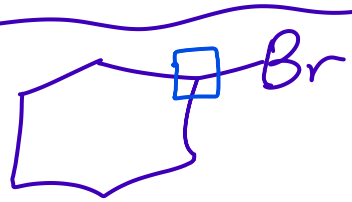
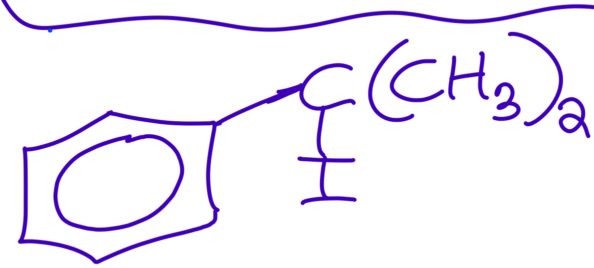
Qn: what is so special about the 9 structures above that makes them different from other structures?

→ Carbon to halogen bond
functional group.
→ Cl
→ Br
→ I

Specific Compound	functional group
$CH_3\boxed{CH_2Cl}$ → $-\overset{\vert}{\underset{\vert}{C}}-Cl$	carbon to chlorine bond
	carbon to bromine bond
	carbon to iodine bond

Research:
→ in nature
→ Occurence
→ Natural
→ Sources
→ Artifical
of alkylhalides.

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Structure ! & $\boxed{\text{IUPAC}}$ Nomenclature

$\boxed{P\,S\,N}$ → Parent (alkane) Name

$\boxed{P}$osition of → $\boxed{\text{Side}}$ chain

(A). CH_4 (methane) + $\boxed{Cl_2}$ → $\boxed{C}H_3Cl$

Brings a chloro-group to

* $\boxed{P}$ S N

$\boxed{\text{chloro}}$methane

[ONE WORD].

(B). CH_3CH_3 ethane + $\boxed{Br_2}$ → CH_3CH_2Br

Brings a bromo group

Bromoethane

$\boxed{P}$ S $\boxed{N}$

CH_3CH_2 with Br (1)

(1) CH_2CH_3 with Br

(C). $\boxed{P}$

(1) $CH_2CH_2CH_3$ with Cl

P S N

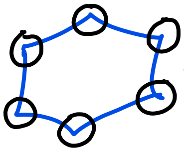
1-chloropropane

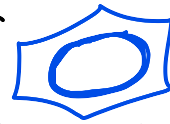
$CH_3CH_2CH_2$ with Cl (1)

SAME

$\overset{1}{C}H_3\overset{2}{C}H_2\overset{3}{C}H_3 + Cl_2$

CH_3CHCH_3 with Cl (2) → $\boxed{2}$-chloropropane.

 → cyclohexane

 → Benzene Ring

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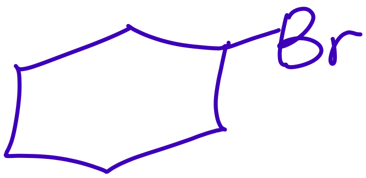
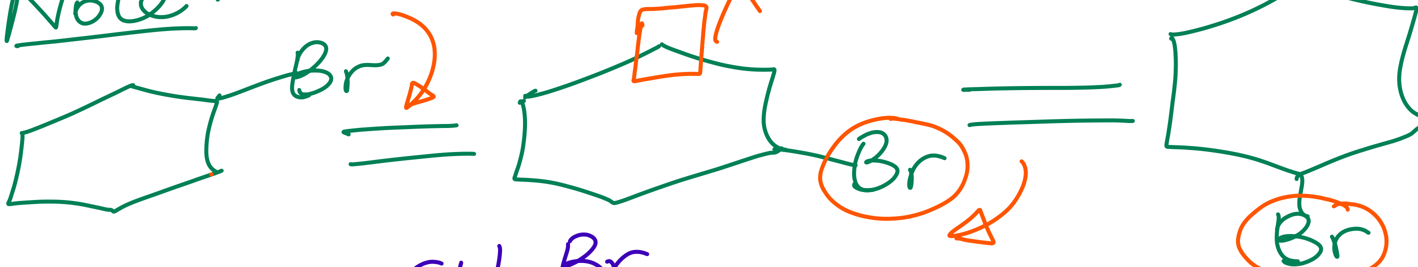


$\neq$

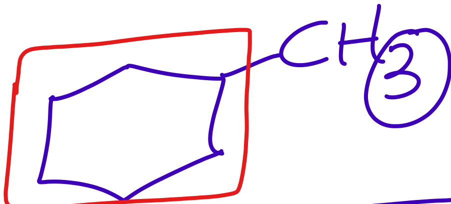
{Bromocyclohexane}

Bromobenzene

Note:



Before;



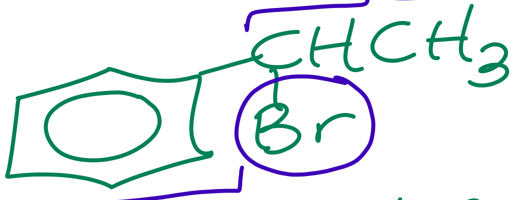
\+ Br_2 → -Bromo

{Methylcyclohexane}

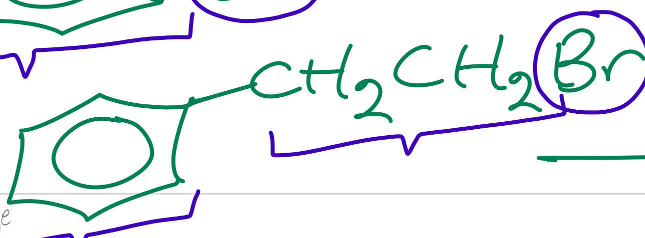
Research: → Help

Nomenclature of aryl halides!

E.g



→



→

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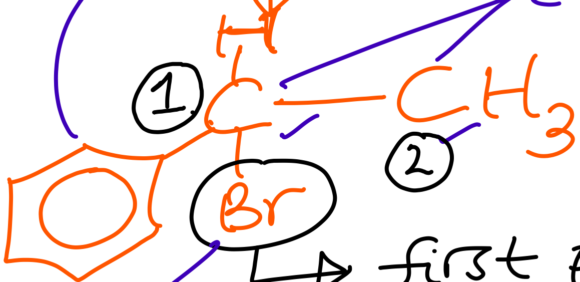
Ethane

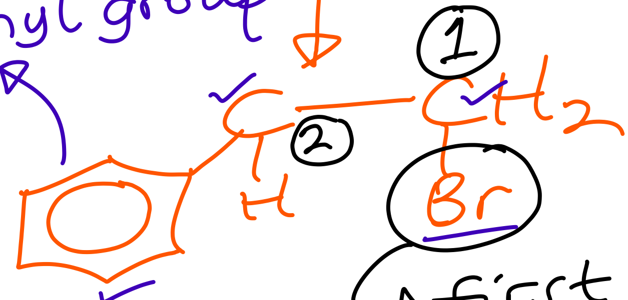
$\boxed{CH_3CH_3}$

Phenyl group

Ethane

phenyl group





first priority

first priority

Bromo

Its the functional group

1-Bromo-1-Phenylethane

1-Bromo-2-phenylethane

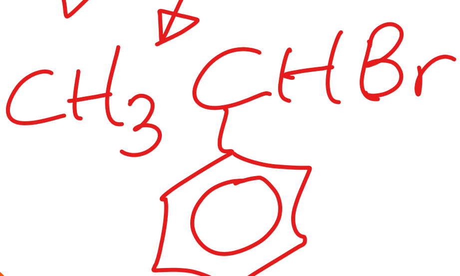
$\boxed{\text{But}}$

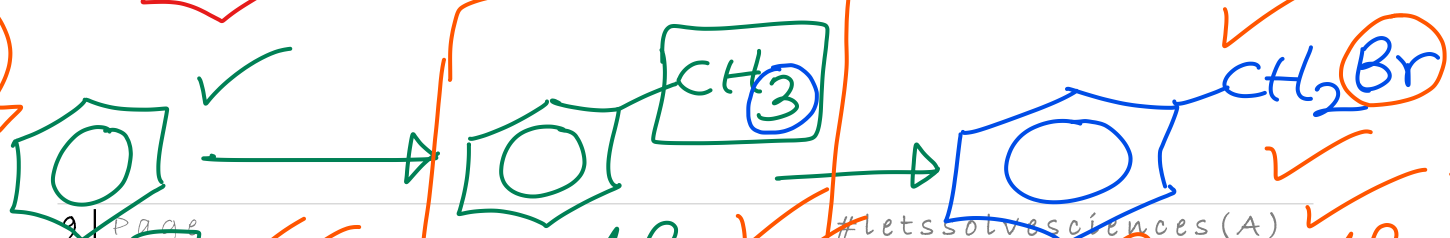
$\boxed{\text{1-Phenyl}}$bromoethane

2-phenylbromoethane

CH_3CH_2Br

Bromoethane





B

MB

BromoMB

Lastly;

$CH_3CH_3 \rightarrow CH_2(Br)CH_2(Br)$

1,2-dibromoethane

P S N

$CH_2(Cl)CH_2(Br)$

A, B, C

$CH_3CH_2CH_3 \rightarrow CH_3CH_2CH(Cl)_2$

1,1-dichloropropane

$CH(Cl)_2CH_2CH_3$

$CH_2(Cl)CH_2CH_2(Cl)$

1,3-dichloropropane

$CH_3C(Cl)_2CH_3$

2,2-dichloropropane

$1+2=3$

$CH_2(Cl)CH(Cl)CH_3$

$2+3=5$

$CH_3CH(Cl)CH_2(Cl)$

$2+1=3$

$2+3=5$

1,2-dichloropropane

(*) The above are isomers of $C_3H_6Cl_2$

(UNEB 2024)
CHEM P1 No 10!

Assignment: Generate the structural formulae of all possible isomers of $C_4H_8Br_2$ and write their IUPAC names.

Physical properties:

(a). Physical states
(b). Solubility
(c). Density

} Make a write-up & present.

(d). Boiling points/Melting Points

OBLIGATORY TASK! → use Internet (AI) & make notes.

→ Roles/Applications/Uses + Ethical considerations of alkylhalides.

Assignment

IUPAC Name

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LASTLY

Directions of the video of Identifying the $\boxed{\text{group}}$ {functional video} → on the drive!

Principal Chemistry folder >> S.5 chemistry folder → Revision & eLibrary → organic chemistry → organic chemistry facilitation → watch the video!

tr Paul Tutor

ScienceScholarsAcademy Ug!

0753279592 {whatsApp calls better}

BYE!